

# Miguel M. G. Pascual-Caballo

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## Summary

PhD student in Fluid Dynamics at ICMAT. My research focuses on the analysis of existence and instability of solutions to partial differential equations in fluid dynamics, using analytical and computer-assisted techniques.

## Education

- 2022 – present    **Instituto de Ciencias Matemáticas (ICMAT)** – Madrid, Spain  
PhD in Mathematics (Fluid dynamics), Universidad Autónoma de Madrid (UAM)  
Advisor: Ángel Castro
- 2021 – 2022    **Universidad Complutense de Madrid** – Madrid, Spain  
Master in Mathematics (Matemáticas Avanzadas) Master's thesis: De Rham theorems in co-homology of manifolds
- 2017 – 2021    **Universidad Complutense de Madrid** – Madrid, Spain  
Bachelor in Mathematics Bachelor's thesis: Geometric mechanics of the rigid body

## Research visits

- 2025    **Brown University** – Providence, RI, USA  
Visiting researcher (3 months), hosted by Javier Gómez-Serrano

## Publications

- 2026    Á. Castro, J. Gómez-Serrano, and M. M. G. Pascual-Caballo. *Linear instability of a Burgers–Hilbert traveling wave*. Preprint, [arXiv:2605.03920](https://arxiv.org/abs/2605.03920)
- 2025    M. M. G. Pascual-Caballo. *Stationary homogeneous solutions for the inviscid SQG equation*. Preprint, [arXiv:2510.03108](https://arxiv.org/abs/2510.03108)

## Talks

- 2026    **Linear instability of a Burgers–Hilbert traveling wave**  
Partial Differential Equations and Applications Seminar, IME-USP, São Paulo, Brazil
- 2025    **A Brief Introduction to Computer-Assisted Techniques in Analysis**  
BYMAT, Sevilla, Spain
- 2024    **Linearization and Stability in Fluid Dynamics**  
BYMAT, Valladolid, Spain
- 2023    **Compactness: Geometry and Topology**  
Junior Colloquium (PhD student seminar), Universidad Autónoma de Madrid

## Teaching

- 2024 – 2025    Teaching assistant, Mathematics I (Biomedical Engineering)  
Universidad Autónoma de Madrid
- 2024 – 2025    Teaching assistant, Algebra (Computer Engineering)  
Universidad Autónoma de Madrid
- 2023 – 2024    Teaching assistant, Calculus II  
Universidad Autónoma de Madrid

## Technical skills

### Programming

Python (SageMath)

### Mathematical software

MATLAB, Mathematica

### Languages

English (Upper Intermediate)

## Honors and scholarships

- 2022    JAE Summer School, ICMAT
- 2017    Bronze Medal, Spanish Mathematical Olympiad